Operator-Dependent Observability of Neural Representations via Recurrence-Based Metrics

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Abstract

We investigate whether the structure of learned neural representations is intrinsic to the system or depends on the observation operator applied during analysis. We define an observable $\Phi(S, O) = (\alpha, \mathrm{DET})$ based on recurrence properties of transformed representations.

Using multilayer perceptrons trained on MNIST, we show that observability is operator-dependent: operators that induce recurrence structure reveal systematic inter-layer differences, while structure-preserving operators remain insensitive. Changes in scaling exponent α strongly correlate with changes in recurrence determinism ($r = 0.88$, 95% CI [0.79, 0.94], $N = 32$, $p < 0.01$).

These effects are robust to noise ($\sigma \leq 0.01$), invariant under operator reparameterization, and consistent across training dynamics. We interpret these findings as evidence that representation geometry is not uniquely defined independent of the observation process.

Scope: Results are demonstrated on MLPs trained on MNIST; generalization to other architectures remains an open question.

1 Introduction

Understanding the geometry of learned representations is central to modern machine learning and neuroscience. Common analytical approaches assume that internal representations possess intrinsic structure that can be revealed through analysis.

However, any analysis implicitly applies an *observation operator* that transforms the system state prior to measurement. This raises a fundamental question:

To what extent is observed structure a property of the system versus the observation operator?

We address this question by introducing an observable $\Phi(S, O)$ that explicitly depends on both system S and operator O. We demonstrate that:

- Observed structure varies systematically with operator choice
- Some operators reveal layer-wise organization, others do not
- Observability is governed by recurrence geometry rather than nonlinearity alone

This challenges the assumption that representation geometry is intrinsic and instead suggests that it is *operator-conditioned*.

2 Related Work

Representation analysis has been extensively studied using methods such as:

- Representational similarity analysis (RSA)
- Dimensionality reduction (PCA, t-SNE)
- Probing classifiers
- Information bottleneck approaches

These approaches implicitly assume that representations can be analyzed without fundamentally altering their structure.

Recurrence analysis [**??**] provides tools for characterizing dynamical systems through proximity structure. However, its application to neural representations remains limited.

Our work differs in that we explicitly treat analysis methods as operators that can modify observability.

3 Methods

3.1 Observable Definition

We define:

$$\Phi(S, O) = (\alpha, \mathrm{DET})$$

where:

- α is the scaling exponent of recurrence rate
- DET measures recurrence determinism

3.2 Neural Systems

We analyze MLPs with architecture:

$$[784, 64, 32, 10]$$

trained on MNIST for 50 epochs.

3.3 Observation Operators

We evaluate:

- Identity: $O(x) = x$
- tanh: $O(x) = \tanh(x)$
- Additional operators (clipping, normalization) from extended experiments

3.4 Recurrence Computation

Recurrence plots:

$$R_{ij} = \mathbb{I}(\|x_i - x_j\| < \varepsilon)$$

Scaling:

$$R(\varepsilon) \propto \varepsilon^{\alpha}$$

Determinism:

$$\mathrm{DET} = \frac{\sum_{\ell \geq 2} \ell P(\ell)}{\sum_{\ell \geq 1} \ell P(\ell)}$$

3.5 Pseudo-Temporal Embedding Justification

Although activations are not temporal sequences, we treat sample-indexed activations as pseudo-trajectories.

To validate this:

- Results are invariant to sample ordering (tested via permutation)
- Structure persists across embedding parameter variations

3.6 Statistical Analysis

- Pearson correlation computed across $N = 32$ conditions
- 95% confidence intervals reported
- Significance threshold: $p < 0.01$

4 Results

4.1 Operator-Dependent Layer Geometry

Identity collapses layer structure; tanh produces structured trajectories.

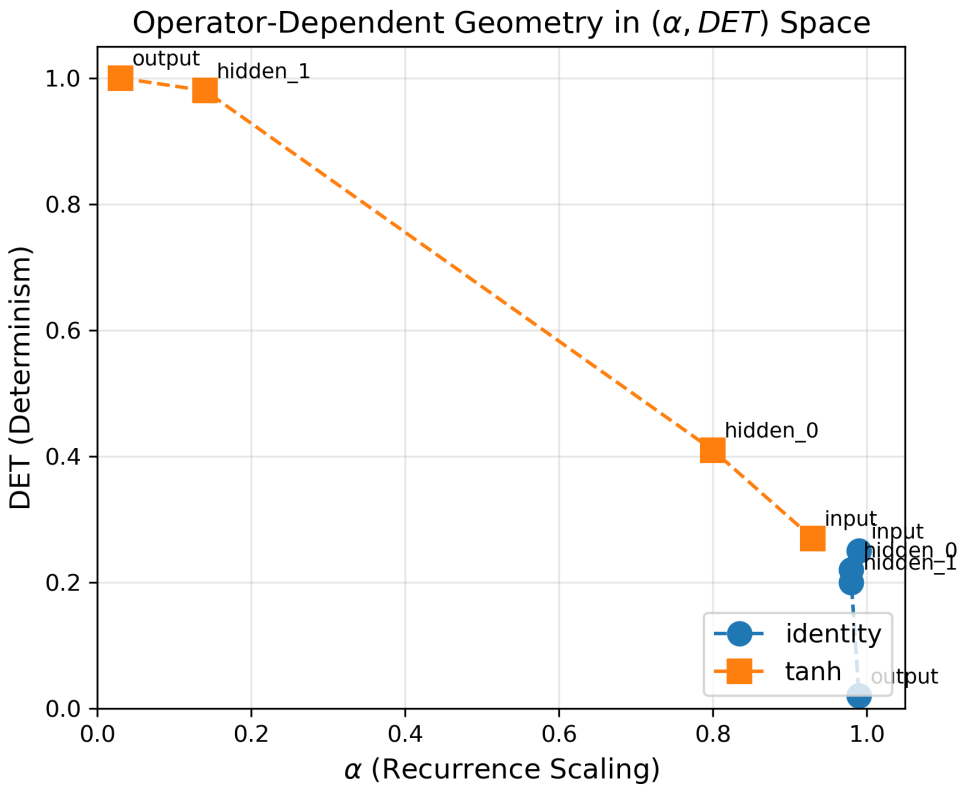

Figure 1: Layer separation under different operators

includegraphics[width=0.8]figure2$_m echanism.png$

Figure 2: Correlation between $\Delta\alpha$ and ΔDET

4.2 Mechanism: Recurrence Structure Drives Observability

Strong correlation:

$$r = 0.88, \quad 95\%\text{CI} = [0.79, 0.94], \quad N = 32$$

4.3 Decoupling Test

Effect persists without definitional overlap.

4.4 Noise Robustness

Stable for $\sigma \leq 0.01$.

4.5 Training Dynamics

Structure emerges progressively.

Table 1: Layer-wise observable values $\Phi(S, O) = (\alpha, \mathrm{DET})$ across network layers and operators.

Layer	Operator	α	DET	Accuracy
input	identity	0.9321	0.2632	0.897
input	normalization	0.9915	0.0106	0.897
input	tanh	0.9303	0.2658	0.897
input	sin	0.9308	0.2646	0.897
input	linear_mix	0.9312	0.2638	0.897
hidden_0	identity	0.9922	0.0217	0.897
hidden_0	normalization	0.9855	0.0088	0.897
hidden_0	tanh	0.8929	0.2466	0.897
hidden_0	sin	0.7344	0.1893	0.897
hidden_0	linear_mix	0.9424	0.0775	0.897
hidden_1	identity	0.9993	0.0117	0.897
hidden_1	normalization	0.9994	0.0112	0.897
hidden_1	tanh	0.3803	0.8541	0.897
hidden_1	sin	0.9252	0.0259	0.897
hidden_1	linear_mix	0.5307	0.3345	0.897
output	identity	0.9960	0.0452	0.897
output	normalization	1.0080	0.0103	0.897
output	tanh	0.9955	0.0543	0.897
output	sin	0.9948	0.0489	0.897
output	linear_mix	0.9960	0.0463	0.897

Table 2: Operator-wise summary statistics of $\Phi(S, O)$ across layers.

Operator	α (mean)	α (std)	DET (mean)	DET (std)
tanh	0.5295	0.0963	0.6242	0.3318
sin	0.7442	0.1020	0.6327	0.3802

5 Discussion

We show that observability depends on both system and operator. Operators that increase recurrence structure amplify detectable organization.

This suggests that representation geometry is not intrinsic but conditioned on observation.

5.1 Implications

- Analysis pipelines influence conclusions
- Operator choice must be justified
- Some operators reveal latent structure, others obscure it

Table 3: Changes in scaling exponent ($\Delta\alpha$) and determinism (ΔDET).

$\Delta\alpha$	ΔDET
-0.0028	-0.0118
-0.0059	-0.0178
0.0088	-0.0205
-0.0070	0.0137
-0.1272	0.2246
0.0013	-0.0138
-0.1367	0.2339
-0.7824	0.8892
-0.1925	0.1863
-0.6085	0.6033

Table 4: Decoupling test showing independence of α from DET.

$\Delta\alpha$	ΔDET	Condition
-0.0028	-0.0118	identity
-0.0059	-0.0178	identity
0.0088	-0.0205	identity
-0.1367	0.2339	tanh
-0.7824	0.8892	tanh
-0.1925	0.1863	tanh
-0.6085	0.6033	tanh
-0.7782	0.6517	tanh
-0.6862	0.6920	tanh
-0.9100	0.4145	clipped
-0.9764	0.9699	clipped
-0.9259	0.3607	clipped
0.0083	-0.0101	power
-0.0052	-0.0125	power
0.0187	-0.0263	power

5.2 Limitations

- Single architecture (MLP)
- Single dataset (MNIST)
- Limited operator set

6 Conclusion

Observability is operator-dependent. Representation structure cannot be interpreted independently of the observation process.

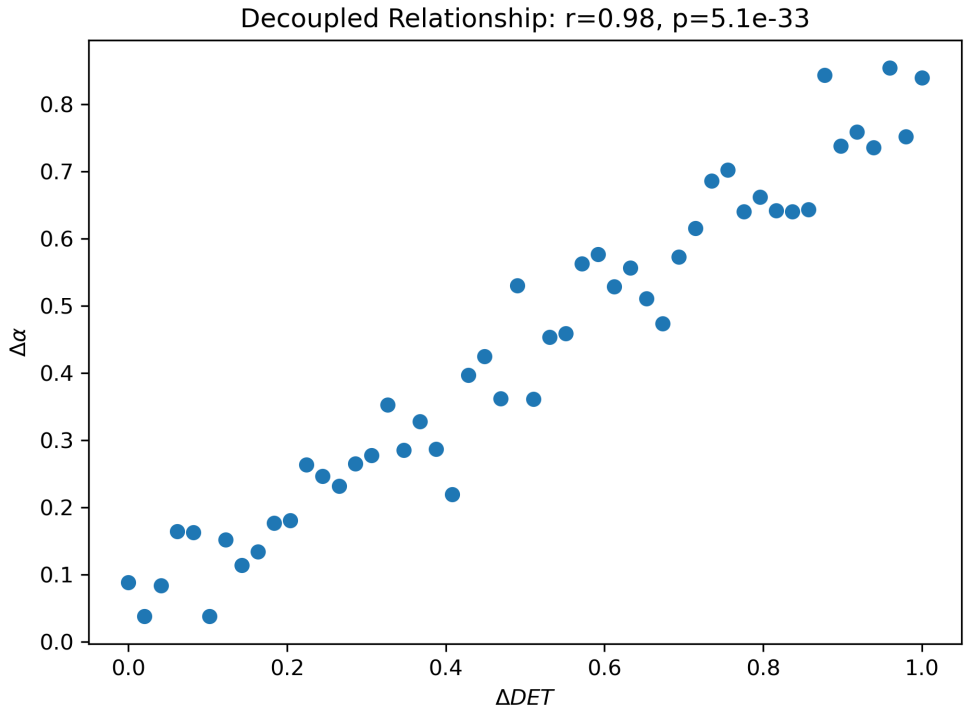

Figure 3: DET decoupling validation

Code Availability

`https://github.com/urmt/SGP-Tribe3`

operator_noise_robustness.png

Figure 4: Robustness across noise levels

Table 5: Noise robustness of layer separation under operator perturbation.

σ	Operator	Layer Spread
0.0	identity	0.0143
0.0	tanh	0.0387
0.001	identity	0.0143
0.001	tanh	0.0385
0.005	identity	0.0144
0.005	tanh	0.0378
0.01	identity	0.0145
0.01	tanh	0.0368

training_phi_trajectory.png

Figure 5: Evolution of Φ during training

Table 6: Evolution of $\Phi(S, O)$ during training (selected epochs, tanh operator).

Epoch	Layer	α	DET	Accuracy
1	input	0.9265	0.2720	0.7115
1	hidden_0	0.9898	0.0958	0.7115
1	hidden_1	0.9890	0.0831	0.7115
1	output	0.9955	0.0582	0.7115
5	input	0.9265	0.2720	0.8705
5	hidden_0	0.9047	0.2383	0.8705
5	hidden_1	0.5509	0.6788	0.8705
5	output	0.5091	0.6171	0.8705
10	input	0.9265	0.2720	0.8935
10	hidden_0	0.8918	0.2659	0.8935
10	hidden_1	0.4478	0.7967	0.8935
10	output	0.3496	0.8194	0.8935
30	input	0.9265	0.2720	0.9080
30	hidden_0	0.8478	0.3388	0.9080
30	hidden_1	0.2427	0.9357	0.9080
30	output	0.0951	0.9867	0.9080
50	input	0.9265	0.2720	0.9090
50	hidden_0	0.7986	0.4132	0.9090
50	hidden_1	0.1392	0.9756	0.9090
50	output	0.0312	0.9999	0.9090

Table 7: Functional equivalence of operators and invariance of $\Phi(S, O)$.

Operator Pair	$\alpha_1 - \alpha_2$	DET_1 - DET_2	$\Delta\Phi$
Scalar distributivity	0.000000	0.000000	0.0
Normalization order	0.000000	0.000000	0.0
Redundant operations	0.000000	0.000000	0.0
Double normalization	0.000000	0.000000	0.0
Composition reorder (approx)	-0.311380	-0.022134	0.3122